

SADEQ ISAAC

PhD ~ Aerospace / Robotics

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SUMMARY

PhD Robotics Engineer with 10+ years of experience in UAVs, Flight Dynamics, and Autonomous Systems. Technical lead in the development of military VTOL and fixed-wing UAV platforms, specializing in autopilot engineering, flight-control systems, avionics integration, autonomous navigation, and GCS software development. Proven track record delivering complex aerospace systems from concept and simulation to flight validation and operational deployment.

EXPERIENCE

2022 – Present



Indra Sistemas, Madrid, Spain

eVTOL & Propulsive VTOL Systems

Indra Sistemas, Madrid, Spain

- Flight Dynamics & Control Engineer for PEM and LINCE eVTOL programs.
- Designed autonomous navigation architecture for GNSS-denied operations.
- Modified PX4/ArduPilot flight-control firmware and core control loops.
- Designed and tuned flight-control algorithms for multi-rotor and VTOL platforms.
- Led flight-test campaigns, controller tuning, and aircraft validation.
- Developed intelligent mission-planning and autonomous inspection software.
- Designed real-time telemetry and flight-data acquisition systems.
- Integrated external CAN avionics through custom autopilot firmware.
- Developed C# plugins for Visionair GCS in collaboration with UAV Navigation.
- Engineered multi-channel communications with PTUs, gimbals, EO/IR payloads, Air-to-Ground systems, and switching antennas.
- Designed Ground Control Station hardware/software architecture from concept through system integration.
- Developed image acquisition and streaming tools for Basler, FLIR, Epsilon and custom camera payloads.
- Developed software extensions for Silvus and Wavenet tactical radio networks.
- Designed firmware and monitoring software for industrial CAN-based electronic modules.
- Developed real-time 3D Primary Flight Displays (PFD) and avionics status interfaces.
- Developed software for real-time control of industrial ECUs and Engine Management Systems (EMS).



Fixed-Wing Systems

- Integrated and configured UAV Navigation VECTOR-600 autopilot for military fixed-wing UAVs.
- Designed resilient Serial/Ethernet avionics networks integrating radios, payloads, antennas, GCS and airborne systems.
- Technical Lead for EUREKA maritime UAV project involving ship tracking and EO/IR gimbal integration.
- Developed advanced Visionair plugins in collaboration with UAV Navigation.

eVTOL & Fixed-Wing Systems / Flight Dynamics / Autopilot Firmware

2023 – 2024



Ahyres S.L., Madrid, Spain

GCS SW for Inspection Missions (Part-time)

Ahyres S.L., Madrid, Spain

- Designed a single-UAV GCS SW for online & offline server missions.
- Designed a QT compact portrait SW to hasten the control of UAV.
- Integrated 2D/3D maps for better terrain monitoring.
- Developed an automated payload view, including HUD, instruments, and video to speed up the mission.
- Integrated online/offline modes to download flight plans automatically from any servers.

GCS SW / QT SW / 2D/3D Maps / Payload View

2020 – 2022



Drone-Hopper S.L., Madrid, Spain

Wild Hopper: Heavy-Duty UAV for Firefighting

Drone-Hopper S.L., Madrid, Spain

- Designed and developed the Wild Hopper prototype, a heavy-duty UAV for day and night firefighting operations.
- Engineered a robust UAV capable of operating in challenging firefighting conditions.
- Collaborated with team to optimize day/night operational efficiency.
- Published in Heliyon (2022) and International Journal of Online and Biomedical Engineering (2021).

Heavy Hybrid-Electric Quadcopter with Flap Control

- Mathematically modeled and designed a heavy hybrid-electric quadcopter controlled by flaps.
- Designed innovative flap-based control system.
- Conducted aerodynamics analysis for hybrid-electric propulsion.
- Published in Unmanned Systems (2022).

Modern GCS for Swarm Missions

- Designed a fast multi-UAV path planning GCS SW for optimal area coverage in aerial sensing applications.
- Adapted algorithms for efficient multi-UAV coordination.
- Developed a QT-based friendly UI for remote sensing operations.
- Developed ROS commands via SSH to handle multi-UAV operations simultaneously.

UAV Firefighting / Hybrid Quadcopter / Swarm Missions

2019 – 2020

CVAR, UPM, Madrid, Spain

Aerostack Autopilot for Quadcopter

CVAR, UPM, Madrid, Spain

- Developed control and guidance systems for an autonomous quadcopter platform, based on Aerostack library.
- Developed control algorithms to stabilize vertical and attitude loops.
- Integrated and developed indoor positioning using stereo cameras.
- Integrated an onboard NVIDIA Jetson Nano to handle online image processing.
- Tested flights on various dynamic platforms at CVAR, UPM.

Aerostack Autopilot / Control Algorithms / NVIDIA Jetson Nano

2012 – 2018

Aerospace & UAV Systems Engineer

PROMAK, PARS, & Academic Research Labs

PROMAK, PARS, & Academic Research Labs

- **GCS & Autopilot Software:** Programmed advanced Ground Control Stations for cartographic UAVs and integrated Piccolo Cloud Cap systems with custom virtual autopilot COMBoxes.
- **Flight Control & Simulation:** Engineered MATLAB Sim-Scape dynamic models for multirotors; compared sliding/adaptive control loops and developed Airbus747 flight dynamics simulations.
- **Computer Vision & Systems Integration:** Deployed real-time image processing pipelines using OpenCV and integrated complex software environments over ROS.
- **Aerodynamic & Structural Design:** Developed Monte Carlo Circular Error Probability (CEP) flight calculators, analyzed wing structures via Abaqus, and executed conceptual commercial aircraft modeling.

Skills: GCS Development / Control Algorithms / MATLAB & Simulink / Image Processing / ROS / Structural Analysis

EDUCATION

Sep 2019 -

PhD in Automation & Robotics Engineering

Oct 2024

 **Specialized in** Flight Dynamics, Control, and Robotics Algorithms.

 **CAR, Polytechnic University of Madrid (UPM), Spain.**

 **Thesis:** "Control a Multi-Ducted Fan UAV Using Thrust Vectoring."

 **Supervisor:** Prof. Pascual Campoy (pascual.campoy@upm.es).

★ **Grade:** Sobresaliente.

Sep 2016 -

MS in Aerospace Engineering

Mar 2019

 **Specialized in** Flight Dynamics and Control.

 **Amir Kabir University of Technology (Polytechnic).**

 **Thesis:** "Simulation and Implementation of the Landing Phase of a Quadcopter on a Moving Platform."

 **Supervisor:** Dr. Naghash (naghash@aut.ac.ir).

★ **GPA:** 3.6.

Sep 2011 -

BS in Aerospace Engineering

Sep 2015

 **Khajeh Nasir University of Technology.**

 **Thesis:** "Autopilot Code Programming for a Flying Robot Using C#."

 **Supervisor:** Prof. Roshanian (roshanian@kntu.ac.ir).

★ **GPA:** 3.5.

PUBLICATIONS

- Feb 2025  **Unmanned Aerial Vehicle-Based Hyperspectral Imaging and Soil Texture Mapping with Robust AI Algorithms**
[Drones, 2025, 9\(2\), 129](#)
P. Flores Peña, **M.S. Ale Isaac**, D. Gifu, E.M. Pechlivani, A.R. Ragab.
- Sep 2024  **Advanced Control Strategies for Securing UAV Systems: A Cyber-Physical Approach**
[Applied System Innovation, 2024, 7\(5\), 83](#)
M.S. Ale Isaac, P. Flores Peña, D. Gifu, A.R. Ragab.
- Jun 2024  **Sensing and Control Integration for Thrust Vectoring in Heavy UAVs: Real-World Implementation and Performance Analysis**
[Unmanned Systems, 2024, 1-23](#)
M.S.A Isaac, P.F. Peña, M.A. Luna, A.R. Ragab, P. Campoy.
- Jun 2023  **Thrust Vectoring Control for Heavy UAVs, Employing a Redundant Communication System**
[Sensors, 2023, 23\(12\), 5561](#)
Ale. Isaac, M. S., Ragab, A. R., Luna, M. A., Ale Eshagh Khoeini, M. M., and Campoy, P.
- Nov 2022  **A Proposed System for Multi-UAVs in Remote Sensing Operations**
[Sensors, 2022, 22\(23\), 9180](#)
P. Flores Peña, Luna, M. A., Ragab, **Ale. Isaac, M. S.**, Ragab, A. R., Elmenshawy, A., Martín Gómez, K., Campoy, P., and Molina, M.
- Sep 2022  **Medium-Scale UAVs: A Practical Control System Considering Aerodynamics Analysis**
[Drones, 2022, 6\(9\), 244](#)
Ale. Isaac, M. S., Luna, M. A., Ragab, A. R., Ale Eshagh Khoeini, M. M., Kalra, R., Campoy, P., and Molina, M.
- Jun 2022  **Wild Hopper: A Heavy-Duty UAV for Day and Night Firefighting Operations**
[Heliyon, 2022, 8\(6\)](#)
Peña, P. F., Ragab, A. R., Luna, **M. A., Isaac, M. S. A.**, & Campoy, P.
- Mar 2022  **Fast Multi-UAV Path Planning for Optimal Area Coverage in Aerial Sensing Applications**
[Sensors, 2022, 22\(6\), 2297](#)
Luna, M. A., **Ale. Isaac, M. S.**, Ragab, A. R., Campoy, P., Flores Peña, P., and Molina, M.
- Jan 2022  **Systemic Integrated Unmanned Aerial System**
[International Journal of Online & Biomedical Engineering \(iJOE\), 2022, 18\(01\)](#)
P., Flores Peña, Ragab, A. R., Luna, M. A., **Ale. Isaac, M. S.**
- Nov 2021  **Mathematical Modeling and Designing a Heavy Hybrid-Electric Quadcopter, Controlled by Flaps**
[Unmanned Systems, 2022, 10\(03\), 241-253](#)
Ale. Isaac, M. S., Ragab, A. R., Garcés, E. C., Luna, M. A., Peña, P. F., and Cervera, P. C.
- Sep 2021  **WILD HOPPER Prototype for Forest Firefighting**
[International Journal of Online and Biomedical Engineering \(iJOE\), 2021, 17\(09\)](#)
Ahmed Refaat Ragab, **Sadeq Isaac**, Marco A. Luna, Pablo Flores Peña.
- Conf.
Jun 2023  **Spiral Coverage Path Planning for Multi-UAV Photovoltaic Panel Inspection Applications**
[2023 International Conference on Unmanned Aircraft Systems \(ICUAS\)](#)
Luna, M.A., **Ale. Isaac, M. S.**, Fernandez-Cortizas, M., Santos, C., Ragab, A.R., Molina, M., and Campoy, P.
- Conf.
Sep 2019  **Control and Guidance of an Autonomous Quadcopter Landing Phase on a Moving Platform**
[11th International Micro Air Vehicle Competition and Conference](#)
Ale. Isaac, M. S., Naghash, A., and Mirtajedini, S.